Shivam Vipul Kapadia

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Education

University of Illinois Urbana-Champaign | *B.S. Computer Engineering, Semiconductor Minor* Expected Dec. 2028

- **Honors:** GPA 3.96/4.00, Dean's List (Fall 2025, Spring 2026), James Scholar, IEEE-HKN
- **Scholarships:** Grainger Engineering Achievement Scholarship, A.R. "Buck" Knight Scholarship
- **Coursework:** Calculus III, Differential Equations, Linear Algebra, Discrete Mathematics, Computer Systems
- **Illinois ECE:** Course Assistant and Grader, ECE 110 Electronics; Grader, ECE 120 Computing
- **Department service:** ECESAC Academic Committee Member; Front Desk Representative

Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, LC-3 Assembly, Bash
Research methods: Comparative analysis, Evidence synthesis, Controlled experiment design, Model evaluation
Systems and data: FastAPI, PostgreSQL, Python asyncio, Supabase, MediaPipe, Tableau, pytest, GitHub Actions, Git

Projects

Meter, LLM Cost Governance Proxy | *Python, FastAPI, PostgreSQL, asyncio* Aug. 2026

- A drop-in **proxy** in front of an AI team's **LLM** calls that meters spend and refuses any request over budget
- Owned the payments and treasury workstream on a four-person, three-day build: **2,000 lines**, **205 tests**
- Designed a **controlled experiment** with a positive control to settle two teammates' opposing reads of a payment API
- Reduced every database read from **201ms to 50ms** by running the **connection pool** in autocommit
- Discovered **CI** had never run the test suites, then repaired **GitHub Actions** and got all **551** checks running

myHR, Multi-Tenant HR Platform | *JavaScript, Supabase, PostgreSQL, MCP* Sep. 2026

- An HR workspace joining an employee directory, onboarding automations and **kanban** boards, with **Slack** sync
- Authored **27** Postgres **Row Level Security** policies enforcing four-role **RBAC**, mirrored in the UI
- Implemented onboarding as a **database trigger**, so adding an employee builds their checklist in one **transaction**
- Integrated AI agents through an **MCP server** with **15 tools**, behind role-scoped, **SHA-256** hashed API keys

Applywise, Grounded Document Generation | *Next.js, TypeScript, Prisma, SQLite* July 2026

- A job tracker that drafts a cover letter, resume bullets and interview prep from a posting using an **LLM**
- Enforced **grounded output** twice, in a system prompt and a server-side stripper, so one prompt failure cannot leak through

PalmSlash, Real-Time Hand Tracking | *JavaScript, MediaPipe, p5.js* June 2026

- A webcam browser game that tracks both hands in real time, running fully **client-side** with no server
- Optimized tracking to **60 fps** by throttling **MediaPipe** to **22 Hz** and predicting between samples with a **One Euro filter**

Experience

Pandit Deendayal Energy University | *Machine Learning Research Assistant* Jan. 2024 – Feb. 2025

- Comparative analyses of **machine learning** for environmental forecasting with Dr. Manan Shah, one in review
- **Air quality:** Compared **SARIMA**, **KNN** and **SVM** across published results; **hybrid models** forecast best
- Published in *Machine Learning for Computational Science and Engineering*, **Springer**, vol. 1, art. 14, 2025
- **Rainfall, first author:** Surveyed **70 papers** on **SVM** and **Random Forest** forecasting, mapping where claims conflict
- Recommended pairing ML with **time-series** data, and proposed **location-based** phone alerts as a flood warning

Revamp, Agentic AI Consulting at UIUC | *Software Engineer* Sep. 2026 – Present

- **Clearline**: a tool turning scanned billing PDFs into reviewed spreadsheets, a job of **6–9 hours** a week by hand
- Structured corrections as an **overlay**, so extracted values, **OCR** confidence and source pages are never overwritten
- Validated **16** schema-derived fields **all-or-nothing** in **FastAPI**, so no bad input half-saves a record
- Engineered **atomic** local saves (temp file, then replace) that restore on reopen, so a crash never loses progress
- Added **59 tests**, including **XSS** payloads, with **CI** green on **Ubuntu** and **Windows**

1Rivet | *Software Engineering Intern* May 2024

- Built the web screens a manufacturing client's customers used, hand-coding the whole front end in **HTML** and **CSS**
- Wrote the **SQL** queries that store and retrieve company records, and ran backup, recovery and purge routines